

KRUPPE: Knowledge-based Research Using Pathway and Procedural Engine

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Abstract

Large Language Models (LLMs) have rapidly evolved from specialized NLP tools to general-purpose models aimed at achieving Artificial General Intelligence (AGI). In light of these developments, this paper explore LLM's capability to answer complex questions and conduct general research, specifically within the finance domain. I present KRUPPE, a framework inspired by a two-decade-old analyst decision model. KRUPPE conducts financial research by dividing the research process into three parts: background exploration, lead generation, hypothesis exploration. This hypothesis-driven approach enables KRUPPE to generate cohesive, narrative-focused reports designed to uncovering the underlying business story behind a query. To evaluate the system, I conducted a head-to-head comparison in which an LLM compared KRUPPE's output against that of baseline models. Results show that KRUPPE reports produce high-quality narratives even when compared to larger models. The full code for KRUPPE is available on GitHub: <https://github.com/CloudyLeopard/senior-thesis>.

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1 Introduction

In less than a decade, language models have evolved from translation tools into powerful reasoning engines. Early AI research used language models to solve specific NLP tasks—transformer-based language models excelled at traditional tasks like relation extraction, text summarization, and sentiment analysis (Zhao et al., 2025; Devlin et al., 2019; Fan et al., 2023). However, researchers discovered these models became even more capable and versatile when *scaled*. Notably, *Large Language Models* (LLM) gain the ability to reason, learn in-context, and follow instructions. This realization sparked a widespread interest in applying LLMs to a range of sectors such as agriculture, business, medicine, and education (Raiaan et al., 2024).

Recent developments signal a shift in LLM research: from solving specific tasks to aiming for more general intelligence. A clear example is using LLMs to pursue Artificial General Intelligence (AGI), or systems with “broad capabilities of intelligence. . . at or above human-level” (Bubeck et al., 2023). Bubeck et al. (2023) even suggests that GPT-4 could be seen as an early form of AGI. Since then, LLMs have grown more powerful, with an explosion of new research models emerging: Deepseek-R1 (DeepSeek-AI et al., 2025), OpenAI’s o1 reasoning model and Deep Research (OpenAI, 2024, 2025a), Claude Sonnet 3.7 (Anthropic, 2025), and Google’s Gemini 2.0 flash thinking (Google, 2025).

Motivated by these developments, I explore LLMs’ potential to conduct financial research—a field notoriously difficult to predict and understand due to complex relationships and the massive amount of information available.

At its core, financial research is trying to make sense of the world. Bouwman et al. (1987) describe this as a two step process: (1) finding relevant information, (2) evaluating the information. Both steps sound simple, but have significant challenges. Analysts face an enormous volume of information today (Schumaker and Chen, 2009), and must have strong reasoning skills to interpret and draw conclusions from that information.

Furthermore, the financial research process is difficult to model or replicate with AI. Since LLMs can process and reason through a large volume of textual data—GPT 4.1’s context size is 1 million tokens, or about 750,000 English words (OpenAI, 2025b)—they seem able to simulate the research process. However, financial markets contain hidden relationships, where events impact seemingly

unrelated entities (Raddant and Kenett, 2021). These relationships are difficult to capture explicitly. Additionally, financial analysis relies heavily on analysts’ intuitions, developed through experience (Bouwman et al., 1987). In interviews, analysts frequently struggle to clearly articulate their decision-making process, often describing it as simply knowing what to look for. Financial research requires more than memorizing information; it requires strong reasoning skills and the ability to make intuitive leaps.

In response, I draw on Bouwman et al. (1987)’s financial analyst decision-making model to structure the financial research process. According to this model, an analyst screens stocks in two phases: “familiarization” and “theme exploration.” The analyst begins by scanning financial documents without a specific goal, aiming primarily to understand the firm and find anything unusual. Once something stands out, they use their intuition—or mental “template” – and form a lead that could explain the situation. From there, research shifts from broad exploration to focused analysis, now with the explicit goal of developing that thesis. I build on this decades-old model by recreating it with modern LLM technology.

I propose **KRUPPE**, or Knowledge-based Research Using Pathway and Procedural Engine. KRUPPE is a research system that answers user questions by prompting an LLM to first develop hypotheses, then conduct hypothesis-driven research. KRUPPE breaks the research process into three stages: background research, lead generation, and hypothesis exploration (Figure 1).

In the background stage, KRUPPE retrieves information only to contextualize the question. In the lead generation stage, the model identifies leads—points of interest that warrant deeper investigation — by reasoning from different “roles” or perspectives. Finally in the hypothesis exploration stage, KRUPPE develops (or rejects) each lead into a hypothesis through a framework I call the RESEARCHTREE. Throughout the process, KRUPPE leverages LLM’s reasoning and tool-using abilities via the ReAct framework (Yao et al., 2023).

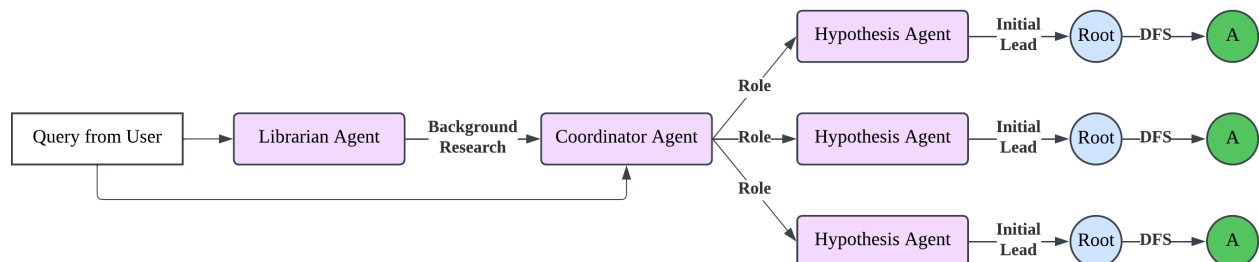


Figure 1: KRUPPE’s Overall Research Model

I evaluate KRUPPE using Microsoft’s method for assessing Graph RAG. I use an LLM-based evaluator to perform head-to-head comparisons across four metrics: comprehensiveness, diversity, empowerment, and cohesiveness (Edge et al., 2024). Test questions are extracted from 73 equity research reports using an LLM. Each question represents a distinct category, varying in content (across firms and industries) and scope (firm-level, industry-level, or macro-level). For each question, the evaluator compares KRUPPE’s output to reports generated by LLMs and traditional equity research reports.

In summary, this study explores the use of LLMs for financial research. Its key contributions are: (1) introducing a new LLM framework by answering financial questions through a hypothesis-driven research approach; (2) evaluating how the system, which uses a smaller model, compares to larger models and whether it remains competitive; 3) examining whether LLM can answer financial queries with the unique insight typically demonstrated by financial analysts.

2 Background & Related Work

2.1 Large Language Models

Language models aim to model “generative likelihood of word sequences” by predicting the probabilities of future tokens (Zhao et al., 2025). Pre-trained Language Models (PLMs), such as BERT, are pre-trained on large text corpora with self-supervised objectives to learn some “general language representations”; the resulting models are then fine-tuned for specific downstream tasks (Devlin et al., 2019). These models have traditionally excelled at natural language tasks like text generation, summarization, translation, question answering, sentiment analysis, and classification.

Large Language Models (LLMs) share the same Transformer architecture and training objectives as smaller PLMs, but are significantly scaled in model size, data volume, and total compute resources (Zhao et al., 2025). This scaling unlocks “emergent abilities”, or capabilities not seen in smaller models (Wei et al., 2022). They include *in-context learning* (learning to solve tasks from examples in a prompt without further training), *instruction following* (performing unseen tasks based on instructions), and *reasoning* (solving complex tasks using techniques like Chain-of-Thought prompting) (Wei et al., 2023; Zhao et al., 2025). These discoveries have shifted research in LLM to-

wards achieving Artificial General Intelligence (AGI), defined as systems that “demonstrate broad capabilities of intelligence... at or above human-level” (Bubeck et al., 2023).

This shift is evident in leading firms’ growing focus on training power reasoning models for general-purpose research. Notable examples include Deepseek AI’s Deepseek-R1 (DeepSeek-AI et al., 2025), OpenAI’s o1 reasoning model and Deep Research (OpenAI, 2024, 2025a), Anthropic’s Claude Sonnet 3.7 (Anthropic, 2025), and Google’s Gemini 2.0 flash thinking (Google, 2025). A standout achievement that highlights this shift is OpenAI’s GPT Deep Research, which scored 26.6 percent on Humanity’s Last Exam—higher than any other reasoning models to date (OpenAI, 2025a). These advancements underscore the motivation behind paper: to explore LLMs’ ability to handle high-level reasoning tasks in a complex domain like financial research.

Another characteristic of LLMs is their ability to take on personas. When prompted to follow specific roles, models have been shown to exhibit stronger reasoning abilities (Kong et al., 2024). Additionally, Brown et al. (2024) found that prompting a model to repeat the same task multiple times improves coverage—the fraction of problems successfully solved. I incorporate both ideas into KRUPPE. KRUPPE prompts the model to generate multiple leads from different perspectives and reason through each one. This encourages exploration of diverse perspectives, hypotheses variety, and raises the likelihood of uncovering the core business story behind the query.

2.2 Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) is a technique that uses LLMs for “knowledge-intensive” tasks. While LLMs can access knowledge from data pre-trained into the model on their own, this ability is limited without an external knowledge source Guu et al. (2020); Lewis et al. (2021). RAG addresses this issue by using input as a query to retrieve relevant contexts from external data. Then, the contexts are appended to the LLM’s prompt. Beyond handling knowledge-intensive tasks, RAG offers two key advantages: (1) flexibility—its retrieval and indexing strategies that can be optimized and fit to different tasks; (2) cost efficiency—retrieving embeddings is much cheaper than processing large contexts through an LLM, making RAG much more economical than deep research models (Gupta et al., 2024).

2.2.1 RAG Limitations

While retrieved contexts can enhance an LLM’s knowledge when answering questions, they also introduce several limitations. Below are a few such limitations:

Response quality depends on context quality: Naive RAG suffers when the retrieved contexts are irrelevant, outdated, or misaligned with the question (Gao et al., 2024)

Coherence challenges: Retrieved contexts should be integrated into a coherent and contextually relevant output. However, combining contexts from different sources can lead to inconsistencies, hallucinations, or misalignment between the contexts and the generated response (Gupta et al., 2024).

Unoriginality: Model may overly rely on the augmented information, generating outputs that only echos information without adding meaningful insights or synthesis (Gao et al., 2024).

Difficulty capturing complex relationships: Naive RAG relies on embedding similarity between the query and candidate contexts, limiting its ability to understand and retrieve information on entities with “intricate relationships”—such as those commonly found in financial markets (Guo et al., 2024).

2.2.2 Attempts to Improve RAG

Previous research has explored many different ways to improve RAG, particularly by improving the quality of the retrieved contexts. Below are a few such approaches relevant to this research:

Modular RAG: A Modular RAG framework attaches specialized components to improve retrieval and processing. These modules can support a range of tasks: a search module to query from different sources, a memory module to leverage LLM’s internal memory for guided retrieval, or a predict module to reduce redundancy by prompting LLM to generate missing information directly.

Structured Data Source: Another approach to improve RAG is to change its data storage strategy. One example is using a knowledge graph instead of traditional vector stores. Knowledge graphs explicitly capture dependencies between entities, helping the system understand relationships and answer higher level, abstract questions more effectively (Guo et al., 2024; Edge et al.,

2024). In other words, GraphRAG can retrieve more contextually coherent information through these relationships, rather than solely relying on embedding similarities. This makes it a promising solution for answering financial queries with complex interdependencies between firms and events. However, building and maintaining GraphRAG’s index is very expensive.

Multi-Step Retrieval: This approach retrieves contexts multiple times by using queries other than the initial input. A single retrieval may not retrieve enough information to adequately answer a complex task. Instead, multi-step retrieval gathers contexts “throughout the generation process” with different queries (Jiang et al., 2023). For example, Jiang et al. (2023) and Su et al. (2024)’s RAG models dynamically trigger retrieval whenever the LLM generates a low-confidence token. Trivedi et al. (2023) combines iterative retrieval with Chain-of-Thought, alternating between retrieval and reasoning until the model arrives at an answer or the chain reaches a certain length. Chen et al. (2024) proposes rewriting the query from multiple perspectives, retrieving contexts for each version.

Chain-of-Thought: LLMs have demonstrated reasoning ability, notably through techniques like Chain-of-Thought where models develop reasoning skills through in-context examples (Wei et al., 2023). Tree-of-Thought extends Chain-of-Thought by having the model generate multiple reasoning paths and choose the most promising one. If the system judges that the current path will fail, it backtracks to a previous reasoning path.

Agentic LLM: Agentic systems enable LLMs to achieve goals by reasoning through the next best action and interacting with external tools. For example, Yao et al. (2023) introduced ReAct, where the model first “reasons” on what to do, then “acts” by invoking the corresponding tool or function.

In this paper, I propose a framework that integrates modular retrieval tools during the lead generation stage, and multi-step retrieval during the hypothesis exploration stage. These methods aim to overcome the limitations of naive RAG. Modular retrieval tools help the LLM synthesize information from different sources. Multi-step retrieval ensures that the retrievals are guided by a central hypothesis. Additionally, KRUPPE’s hypothesis exploration stage introduces RESEARCHTREE, which combines Tree-of-Thought and ReAct to use the strength of both methods: real world interaction through tools, and flexible reasoning that can backtrack when there is insufficient evidence. Together, these approaches ensure that the retrieved context is both cohesive and capable of capturing complex relationships.

2.3 Financial Research

Financial research broadly falls into two categories: technical and fundamental analysis. Technical analysis examines historical price data to forecast future trends. Fundamental analysis evaluates financial statements, company performance, and broader economic indicators to determine investment potential (Schumaker and Chen, 2009; Bauman, 1996). This research primarily focuses on fundamental analysis because LLMs are much better suited to process natural language, and fundamental analysis contains much more qualitative component.

2.3.1 Challenges in Financial Research

Bouwman et al. (1987) describes financial research as a two-step process: 1. analysts need to get the right information, and 2. analysts need to evaluate the information. Furthermore, in the real world, there is an enormous amount of textual data available. Analysts review a variety of sources, from company generated financial documents to independently generated sources such as news outlets, analyst reports, and discussion boards (Schumaker and Chen, 2009). As such, applying LLMs to financial research seems intuitive, given its ability to efficiently process massive amounts of text data. For example, OpenAI’s newest model (GPT-4.1) has a context size of 1 million tokens, or 750 thousand English words (OpenAI, 2025b). However, the task is made more difficult due to the following challenges in financial research:

Complex Relationships in Finance Environment: The financial market often contains hidden dependencies, where an event can have consequences to seemingly unrelated entities (Raddant and Kenett, 2021). An event that appears unrelated from a surface level analysis, may in fact be the root cause of a firm’s performance change.

Dynamic Market and Timing: the market is constantly evolving with new information. A robust analysis system needs to be able to draw upon new information, something that standard LLMs or naive RAG systems are not equipped to do so.

Expert Knowledge: Bouwman et al. (1987) introduces the idea of “expert knowledge” and “financial template”. In simpler terms, financial analysts rely heavily on experience and intuition to guide their research decisions. For example, when they see certain observations, they intuitively can

recognize the situation and generate new hypotheses or leads. This aligns with my own findings—when asked about their research style, analysts often respond vaguely, saying things like “I just know”. Additionally, the idea of “research philosophy” and “creativity” was brought up—each analyst tends to emphasize different preferred qualities when screening a firm, and often draws deeper insights by connecting seemingly unrelated sources of information.

2.3.2 Decision Making Model

This research draws upon Bouwman et al. (1987)’s structured model of financial analysis (Figure 2). His model provides a systematized framework for understanding financial analysts’ decision making process. The model can be divided into two stages: an exploratory phase where analysts familiarize themselves with the company, followed by a theme-driven phase where analysts focus on developing and testing hypotheses.

Phase 1: Familiarization. (Bouwman et al., 1987) describes the first stage of the process—the “familiarizing phase”—as exploratory. At this point, analysts typically have no clear objective other than “looking for something extraordinary” or “new”. The goal is to uncover leads or themes may be worth further investigation. Analysts work through a metaphorical checklist to search for specific information that helps them screen a stock. These checklists fall into two types: a *content-based* checklist, which focuses on reports, sections, and key figures to review, and a *conditional* checklist, which includes information the analyst specifically wants to learn specifically but is not immediately provided in standard reports. Notably, financial statements are the most frequently used document during this phase (Bouwman et al., 1987). This phase continues as long as the analyst have more they want to find out about the firm, or there are items left on the checklist.

Phase 1.5: Lead Generation. The transition from “familiarization” to the “theme exploration” occurs when the analyst generates leads—initial hypotheses or ideas drawn from earlier observations. Bouwman et al. (1987) explains that an analyst identifies these leads through a “template,” a mental model that helps them recognize patterns and form intuitive hypotheses. For example, during the exploratory phase, an analyst may observe that a stock is performing well despite weak earnings and a vulnerable economy. The analyst may intuitively infer that the market anticipates significant improvements from the firm, and use that recognition as a lead for further investigation.

Part 2: Theme Exploration. Bouwman et al. (1987) describes the final stage as “theme” or “hypothesis”-driven, in contrast to the earlier exploratory phase. Once an analyst identifies a lead or a hypothesis from the “familiarizing phase”, the research shifts to developing the hypothesis. In other words, instead of exploring to find new information, analysts now actively seeking information that either supports or rejects the hypothesis. Content wise, Bouwman et al. (1987) noted that analysts move away from financial statements, instead look for more strategic and operational information such as “segments, products, and markets”.

KRUPPE is a system inspired by Bouwman et al. (1987)’s decision making model. It breaks up the research process into the three steps described above, using structured reasoning and information retrieval to replicate the financial analyst behavior.

2.3.3 Other LLM & Finance Research

There are many research that attempts to leverage LLMs specifically within financial contexts. For example, Koa et al. (2024) explored teaching LLMs to predict stock movements by continuously refining forecasts based on prior outcomes. Pei et al. (2024) used LLMs to extract business risk information from financial news. These studies highlight the growing role of LLMs in finance-specific tasks. Additionally, they demonstrate different ways LLM can become involved in the finance industry. KRUPPE builds on these ideas by incorporating them as tools the system can call on to perform analysis—such as interpreting financial statements—within the overall hypothesis-driven research framework.

2.4 Evaluating LLM

Standard evaluation methods for RAG typically involve solving tasks like multi-hop question-answering (QA), and use benchmark datasets such as HotpotQA or GraphQA to assess the AI system’s generated outputs (Gupta et al., 2024). However, these evaluations primarily assess factual accuracy, making them less appropriate for this research’s focus on complex, abstract financial queries that may not have a single correct answer.

A more suitable approach is the evaluation methodology proposed by Edge et al. (2024) for

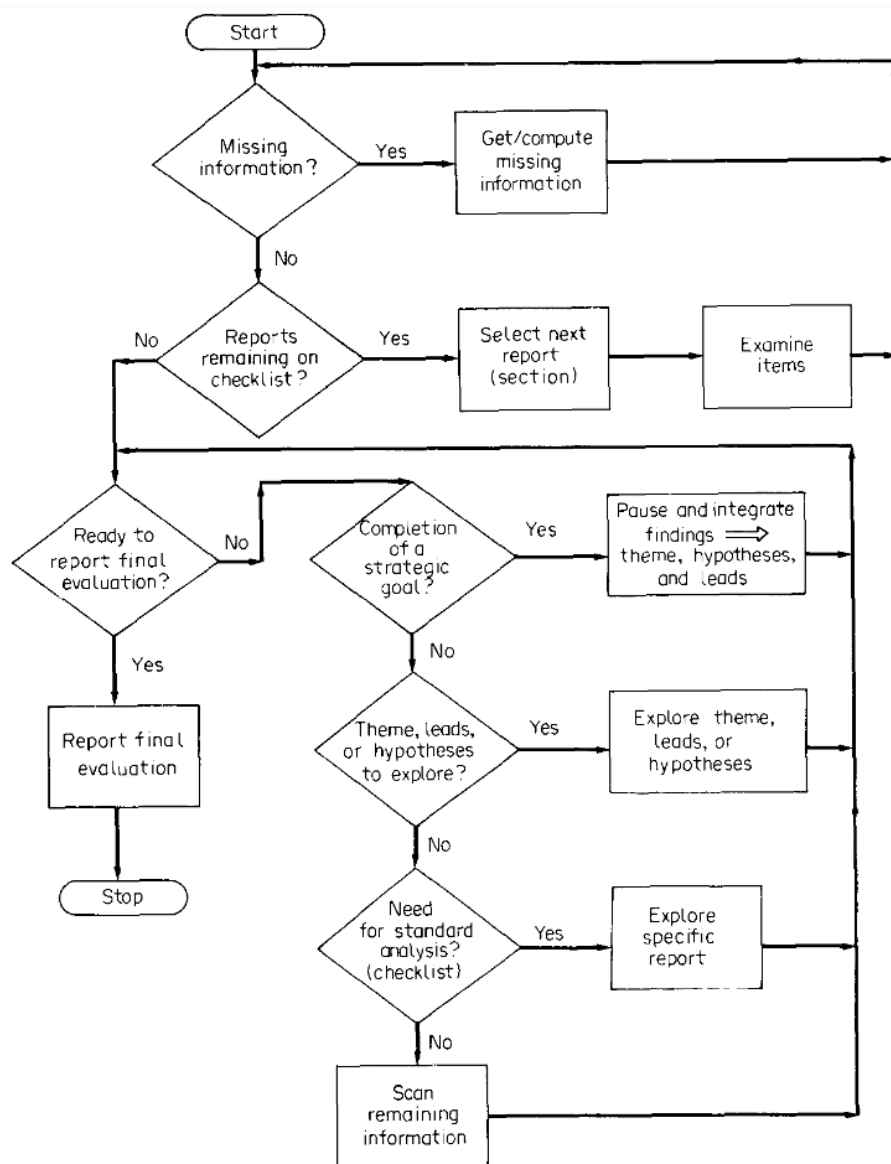


Figure 2: An analyst’s decision making model by Bouwman et al. (1987). The model is broken down into two phases: a “familiarizing phase” where the analyst learns about the firm, followed by “theme exploration” where the analyst researches to develop a specific lead/theme/hypothesis

GraphRAG. Unlike most RAG systems that focus on factual retrieval, GraphRAG’s is designed for “global sensemaking tasks,” which aligns more closely with KRUPPE’s objectives. To evaluate their system, Edge et al. (2024) first prompted an LLM to generate test questions from on a scraped dataset over which GraphRAG will build an index. These questions are then passed through both GraphRAG and baseline models. Finally, an LLM is used as a judge to conduct head-to-head comparisons between the outputs of GraphRAG and the baseline models, using a set of defined qualitative metrics.

3 Methodology

I propose KRUPPE, an LLM framework for financial research that simulates the analyst decision-making model proposed by Bouwman et al.. KRUPPE divides research into three phases: **background research**, **lead generation**, and **hypothesis exploration**. Algorithm 1 describes KRUPPE’s algorithm and Figure 1 visualizes KRUPPE’s overall research flow, both on a very high level.

KRUPPE begins with a user submitted query. First, the LIBRARIAN agent gathers general background information to contextualize the query. Next, the COORDINATOR agent creates several expert roles—each offering a different perspective on the question. These roles are given to HYPOTHESIS RESEARCHER agents, who use their “expertise” and the background research to propose an initial hypothesis.

Each HYPOTHESIS RESEARCHER conducts hypothesis-driven research using a structured method that I call RESEARCH-TREE. This framework iteratively collects evidence and develops the RESEARCHER’s initial hypothesis by strengthening or rejecting it. Once complete, each textscResearcher writes a report to explain their findings.

For every query, KRUPPE produces multiple reports, each built around a different narrative. Some reports support their researcher’s hypothesis, while others may reject it. Together, these reports give the user a range of perspectives, each built around a different story.

Algorithm 1 KRUPPE Query Overall Research Process

Require: User query Q

Ensure: Final answer $\mathcal{A}$

```
1: /* Step 1: Background research */
2:  $BG \leftarrow \text{LIBRARIANAGENT}(Q)$ 
3: /* Step 2: Role generation */
4:  $R \leftarrow \text{COORDINATORAGENT}(Q, BG)$  {Assign each HypothesisAgent a role}
5: /* Step 3: Hypothesis generation and exploration */
6: for all agent  $h$  in  $R$  do
7:    $root_h \leftarrow h.\text{INITIALLEAD}$ 
8:    $A_h \leftarrow \text{DFS}(root_h)$ 
9: end for
10: /* Step 4: Collect answers */
11:  $\mathcal{A} \leftarrow \{A_h \mid h \in R\}$ 
12: return  $\mathcal{A}$ 
```

3.1 Ideal Query

The process begins with a user query Q . Ideally, Q is a business question that explores market dynamics and relationships.¹ Such questions seek to interpret the world through the lens of a specific business entity—a firm, industry, or country. KRUPPE is designed to generate responses built around a central narrative, connecting events and situations to a broader business story.

3.2 Librarian Agent

The LIBRARIAN agent uses a ReAct framework to conduct background research, collecting information that subsequent agents can use to develop potential hypotheses. Bouwman et al. (1987) describes this as a “familiarizing phase,” where analysts methodically work through a “checklist” of items—such as annual reports or key metrics like “firm size” and “divisional profit margins”—to screen a stock. In this phase, analysts often have no specific goals beyond “looking for something extraordinary” or “new.” This broad exploration helps analysts identify themes that are worth investigating further.

KRUPPE emulates this research strategy for query familiarization using the ReAct framework (Yao et al., 2023), which guides the model to reason through what information it needs and then retrieves

¹While KRUPPE is not designed for fact-based queries, it can answer them if the necessary information is accessible through its tools.

it. Algorithm 2 and Figure 3 illustrate this process. From the query, KRUPPE reasons about what background information needs to be collected first. It then *acts* by determining the tool and input to use, and invoking that tool.² The retrieved information is fed back into the message history chain H , which KRUPPE uses to *reason* about the next needed piece of information.³ KRUPPE continues this reasoning-acting loop until it judges that it has gathered enough context to move forward.

At the end, KRUPPE compiles the collected information into a report R . Here, the LLM engine is prompted not to answer the query directly, but instead to provide background context that supports later lead generation task.

This research process mirrors the analysts’ checklist-based approach described earlier. While it may seem that analysts follow a fixed list, Bouwman et al. (1987) writes the process is very intuition-driven. In fact, he writes that analysts “know exactly what [they] want”. As such, KRUPPE lets the LLM decide the information it needs to retrieve based on the query. Like the analyst, it searches for details that help familiarize and make sense of the task. The core idea remains the same: this is an exploratory stage with no fixed goal beyond finding something potentially useful or interesting.

²KRUPPE provides the LLM engine with information about the tools it can access in both the *reason* and *act* step. This way, the LLM knows what type of information it has access to during the *reason* step.

³Newly collected information is stored in a vector storage. This allows the model to retrieve it later using semantic search, which is much faster and cheaper than re-querying and processing external sources.

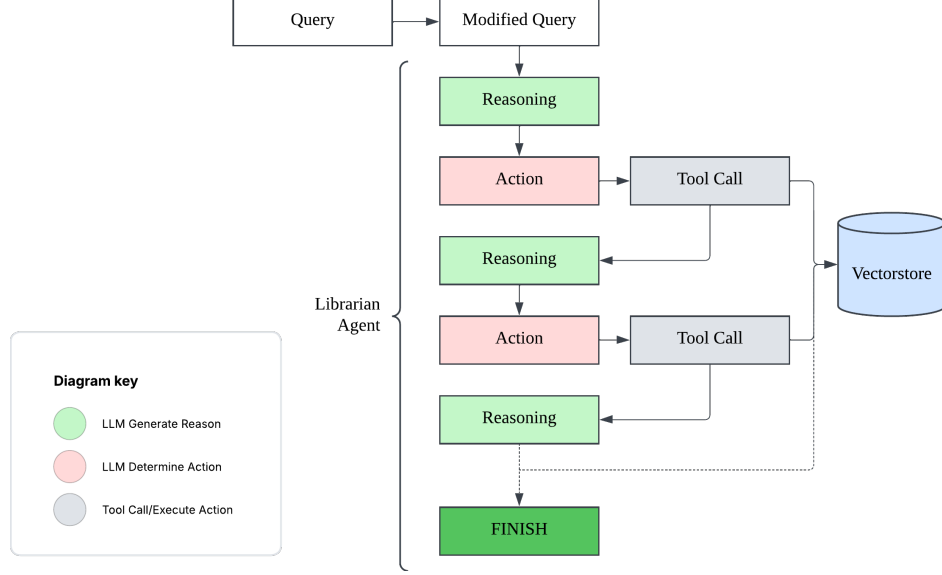


Figure 3: KRUPPE Librarian Model

Algorithm 2 Librarian Agent

Require: User query Q

Ensure: Final report R

```

1:  $Q' \leftarrow \text{MODIFYQUERY}(Q)$ 
2:  $V \leftarrow \emptyset$  {vectorstore}
3:  $H \leftarrow [\text{Message}(\text{user}, Q')] \text{ {chat history}}$ 
4: repeat
5:    $r, a \leftarrow \text{REACT}(H)$ 
6:    $H.\text{append}(\text{Message}(\text{assistant}, r))$ 
7:    $H.\text{append}(\text{Message}(\text{assistant}, a))$ 
8:    $t \leftarrow \text{DECIDEACTION}(a)$ 
9:    $H.\text{append}(\text{Message}(\text{assistant}, t))$ 
10:  if  $t \neq \text{FINISHED}$  then
11:     $d \leftarrow \text{TOOLCALL}(t)$ 
12:     $H.\text{append}(\text{Message}(\text{tool}, d))$ 
13:     $V \leftarrow V \cup \{d\}$ 
14:  end if
15: until  $t = \text{FINISHED}$ 
16:  $R \leftarrow \text{COMPILEREPORT}(H)$ 
17: return  $R$ 

```

3.3 Coordinator

The COORDINATOR agent prepares KRUPPE for the lead generation stage and hypothesis exploration stage by creating domain experts to approach the query from different angles. Algorithm 3

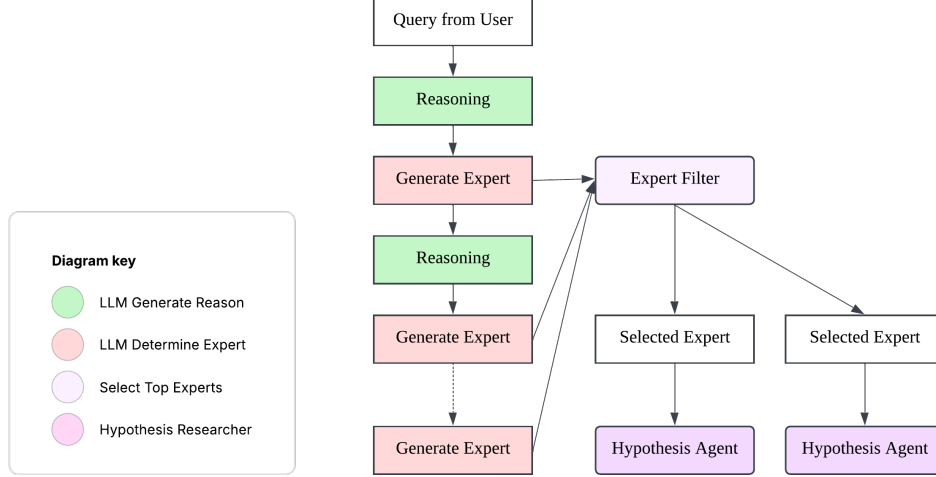


Figure 4: KRUPPE Coordinator Model

and Figure 3 illustrate this process. KRUPPE iteratively identifies experts who offer distinct perspectives on the query Q , prompting each new expert e to differ as much as possible from the previously generated experts. Then, KRUPPE selects the top n most relevant experts to manage cost.⁴ The COORDINATOR initializes a HYPOTHESIS RESEARCHER for each expert by assigning that expert’s role and description.

Assigning domain roles allows KRUPPE to explore different hypotheses and narratives. Kong et al. (2024) shows that role-based prompting improves LLM reasoning in both Zero Shot and Chain-of-Thought settings. From an academic standpoint, having a diverse team where each member can provide their own unique insights and perspectives helps encourage and improve problem-solving (Hershey H. Friedman et al., 2016). A domain role encourages the model to consider different business stories beyond a generic finance approach. For example, in a question about tariffs, an Economist may focus on policy impacts, while a Geopolitical Risk Analyst may instead highlight political tensions—leading to different insights.

Additionally, as KRUPPE generates and explores more hypotheses, the chances of discovering a strong business narrative increase. Brown et al. (2024) showed that while a single LLM output may be weak, performance improves with more samples: “coverage improves through models generating correct solutions to problems they have not previously solved.” By exploring multiple hypotheses, KRUPPE increases the likelihood that at least one narrative captures the core of the issue.

⁴ n is a user-defined parameter that controls the number of perspectives generated.

Algorithm 3 Coordinator

Require: User query Q **Ensure:** List of hypothesis researchers $\mathcal{H}$

```
1: /* 1: Gather expert candidates */
2:  $E \leftarrow \emptyset$ 
3: repeat
4:    $\rho \leftarrow \text{REASON}(Q, E)$ 
5:    $e \leftarrow \text{GENERATE\_EXPERT}(\rho)$ 
6:    $E \leftarrow E \cup \{e\}$ 
7: until No more experts by LLM
8: /* 2: Filter down to the best experts */
9:  $S \leftarrow \text{EXPERT\_FILTER}(E)$ 
10: /* 3: Instantiate a hypothesis researcher per expert */
11:  $\mathcal{H} \leftarrow \emptyset$ 
12: for all  $e \in S$  do
13:    $h \leftarrow \text{HYPOTHESIS\_AGENT}(e)$ 
14:    $\mathcal{H} \leftarrow \mathcal{H} \cup \{h\}$ 
15: end for
16: return  $\mathcal{H}$ 
```

3.4 Hypothesis Researcher

The HYPOTHESIS RESEARCHER builds on the LIBRARIAN’s background research and COORDINATOR’s domain role to answer the user query Q through a hypothesis-driven approach. Bouwman et al. (1987) characterizes this second stage as more “theme” or “hypothesis”-driven. Whereas the earlier familiarizing phase is exploratory, this stage focuses on investigating a specific hypothesis developed from the initial familiarization and research.

After being initialized by the COORDINATOR, each HYPOTHESIS RESEARCHER forms a lead or a working hypothesis which it then develops or rejects. This process is illustrated in Algorithm 4 and Figure 5. Using its assigned domain role e and the background research R , the RESEARCHER constructs a working hypothesis in response to the user query Q . These hypotheses often begin as points of interest from background report R that warrant more research, such as unusual observations that match a specific working hypothesis. It then develops this hypothesis using a method I call RESEARCH-TREE, which combines ReAct, Tree-of-Thought, and Depth-First Search. Once the hypothesis is fully developed, the RESEARCHER generates a report based on its findings, centered around the hypothesis.

The lead generation forces the model to reason creatively, identifying possible business stories from the collected observations through the lens of a unique domain. This is important because financial research often draws on analyst’s intuition. Bouwman et al. (1987) calls this a “template”—a mental model that lets analysts quickly recognize industry patterns and form hypotheses. He gave the following example: if earnings and the economy are weak but a stock is rising, an analyst might intuitively infer that the market expects a rebound. The analyst can then continue research by developing on this lead.

For KRUPPE to effectively mimic this behavior, it must generate distinct leads that can develop into business narratives addressing the user query Q . By assigning different domain perspectives, KRUPPE is pushed to explore connections between different angles, one of which may uncover the core story behind the query.

Additionally, I believe generating a hypothesis before retrieving evidence encourages creativity and improves transparency. Gupta et al. (2024) identifies two key issues in traditional RAG systems: (1) models may over-rely on retrieved content, passively accepting it instead of forming original insights, and (2) RAG systems operate as black boxes, offering little visibility on how retrieval affects generation. By generating the hypothesis first, KRUPPE ensures that future retrieval and generation is anchored on a center idea, increasing transparency and reducing bias toward retrieved content.

With a working hypothesis, the model develops it using a RESEARCH-TREE approach, illustrated in Algorithm 4. The process begins by treating the hypothesis as the root node in a Depth-First Search (DFS). The model reasons multiple times about the current situation and the next action to develop the hypothesis. It chooses the most relevant option and acts by calling on the corresponding tool. The result is added back into the context, and reasoning continues.

This cycle repeats until the model determines that the current research path is unproductive—i.e., there’s not enough information on this research path to support or reject the hypothesis. At that point, it backtracks and explores the next most promising path. Once the hypothesis is fully developed or rejected, the model generates a report summarizing its findings and supporting evidence centered around the hypothesis.

While I have not fully verified the following claims in this paper, I offer the following specula-

tions. Primarily, this research methodology mimics how humans approach research: hypothesis development is iterative, and different reasoning paths can lead to different results. If a path fails, the model tries the next best option. Using ReAct at each step ensures the model’s reasoning is grounded in real-world information at each step. It also enables the model to validate or reject its hypothesis based on whether its expectation is reflected in the real world. Finally, because every step in the research process builds around a central working hypothesis, the final report is expected to form a cohesive narrative, where the retrieved information is all aligned in a central business story that addresses the query.

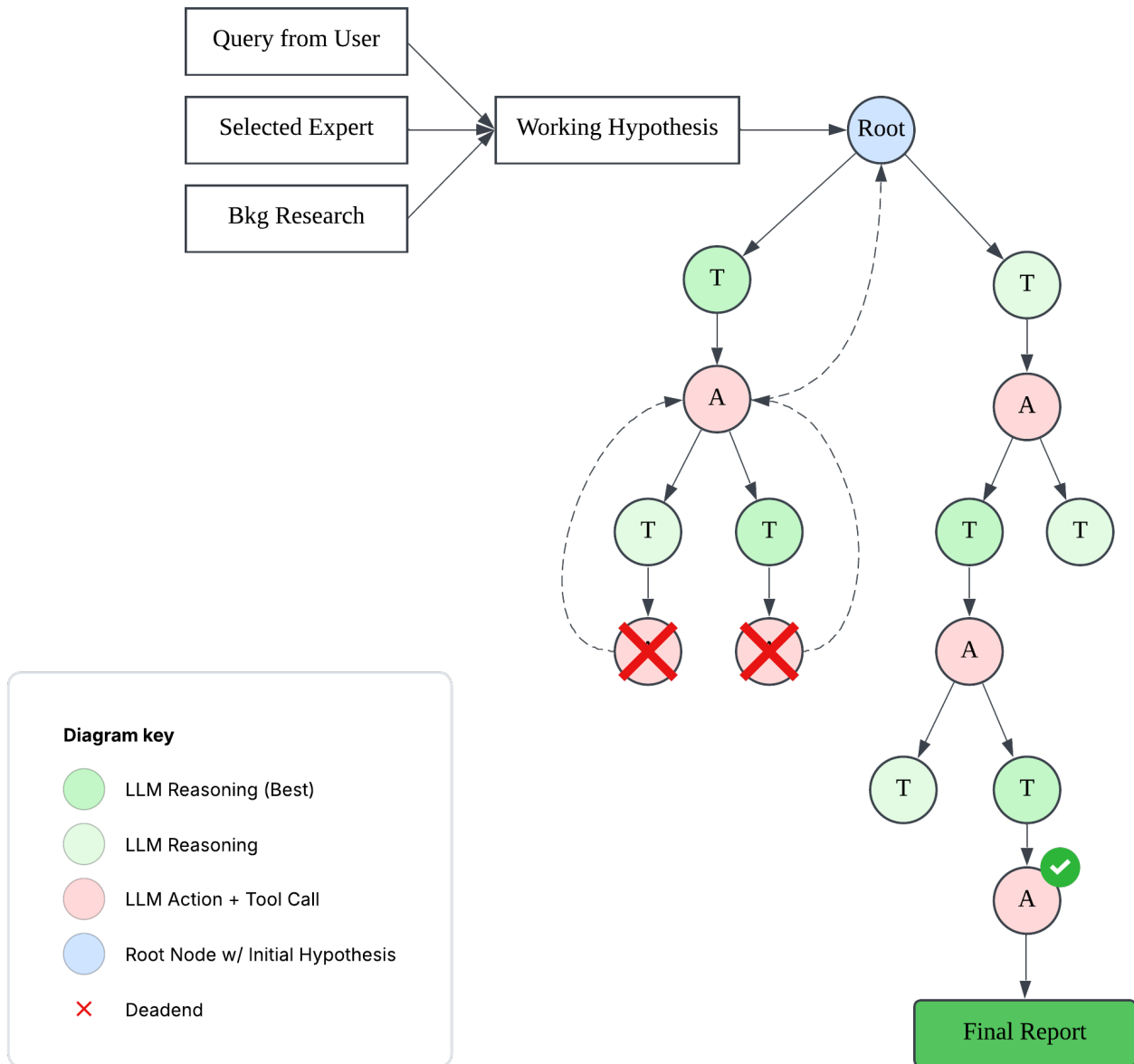


Figure 5: KRUPPE Hypothesis Research (Tree) Model

Algorithm 4 DFS Research-Tree for Hypothesis Development

Require: Query Q , Expert role e , Background research R

Ensure: Final report R_f centered on the hypothesis

```
1: /* 1. Build the working hypothesis */
2:  $H \leftarrow \text{CONSTRUCTWORKINGHYPOTHESIS}(Q, e, R)$ 
3: /* 2. Initialize stack for DFS */
4:  $S \leftarrow [H]$ 
5: while  $S \neq \emptyset$  do
6:    $n \leftarrow S.\text{PEEK}()$ 
7:   if  $n.\text{status} = \text{UNDISCOVERED}$  then
8:      $n.\text{status} \leftarrow \text{DISCOVERED}$ 
9:     if  $n.\text{isLeaf}$  then
10:      if  $\text{ACCEPTHYPOTHESIS}(n)$  then
11:         $R_f \leftarrow \text{GENERATEREPORT}(n)$ 
12:
13:      return  $R_f$ 
14:    else
15:       $n.\text{status} \leftarrow \text{FINISHED}$ 
16:      continue
17:    end if
18:  end if
19:  if  $n.\text{actQueued}$  then
20:     $(m, a) \leftarrow \text{TOOLCALL}(n)$ 
21:     $n.\text{messages}.\text{APPEND}(m)$ 
22:     $n.\text{actResults} \leftarrow a$ 
23:  end if
24:  /* generate and rank child thoughts */
25:   $C \leftarrow \text{GENERATEANDRANKCHILDREN}(n)$ 
26:  for all  $\text{child} \in \text{REVERSE}(C)$  do
27:     $S.\text{PUSH}(\text{child})$ 
28:  end for
29: else
30:    $n.\text{status} \leftarrow \text{FINISHED}$ 
31:    $S.\text{POP}()$ 
32: end if
33: end while
34: /* 3. No branch accepted—compile rejection report */
35:  $R_f \leftarrow \text{GENERATEREJECTIONREPORT}(H)$ 
36: return  $R_f$ 
```

4 Experiments

I evaluate KRUPPE on business queries that explore market dynamics and relationships. To create the test set, I collected 73 equity research reports across various industries, where some are industry wide and others firm-specific. For each report, I used an LLM to extract the core question the report aims to answer. KRUPPE is then tested on these 73 questions and compared against baseline models.

4.1 Model Configuration

KRUPPE uses GPT-4.1-mini, and equipped with the following tools:

- RAG Query: Retrieves previously gathered documents from the vectorstore.
- LLM Query: Consult an external LLM for general knowledge
- News Hub: Collects news articles from Financial Times, New York Times, and News API through the following sub-tools:
 - Search News: Retrieves articles based on keyword queries.
 - Recent News: Retrieves the latest news articles.
 - Archived News: Retrieves articles from a specified time range.
- Fin Hub: Collects financial information through the following subtools:
 - Company Background: Provides a brief summary of a company’s operations.
 - Financial Statements: Returns company financials via yfinance.
 - Analyze: Uses an LLM to interpret and analyze the financial statements.

4.2 Baselines

KRUPPE is compared against the following models’ output

- GPT 4o-mini with Web Search (gpt-4o-mini-search-preview-2025-03-11)
- GPT 4o with Web Search (gpt-4o-search-preview-2025-03-11)
- GPT 4.1-mini (gpt-4.1-mini-2025-04-14)
- GPT 4.1 (gpt-4.1-2025-04-14)

- o3 (o3-2025-04-16)
- Equity Research Report⁵

I also ask analysts to compare and evaluate the results generated by KRUPPE.

4.3 Evaluation Metrics and Methods

The outputs are evaluated using a head-to-head comparison approach, inspired by the method in Edge et al. (2024). For each test question, KRUPPE generates approximately six reports—three distinct perspectives from the COORDINATOR, each creating two reports. Each report is compared to the baseline model’s output using an LLM-based judge. The judge evaluates which report is better, based on the following four metrics:

- **Empowerment:** How well does the answer help the reader understand and make informed judgements about the topic?
- **Cohesiveness:** How well does the answer flow and connect different parts of the information together? Does the answer follow one central narrative?
- **Comprehensiveness:** How much detail does the answer provide to cover all aspects and details of the question?
- **Diversity:** How varied and rich is the answer in providing different perspectives and insights on the question? Does it address the answer from multiple perspectives?

Finally, the LLM judge determines which report is the overall winner.

4.4 Results

I calculated two statistics: (1) the percentage of wins each KRUPPE generated report has over the baseline models, and (2) the percentage of wins that any KRUPPE generated reports in the same test question has over the baseline model. For each percentage win, I computed a 95% confidence interval, graphed the result in bar graphs (Figure 6 and 7), and calculated p-values for statistical significance (Table 1 and 2).

⁵Compared against the original equity research report that generated the test question.

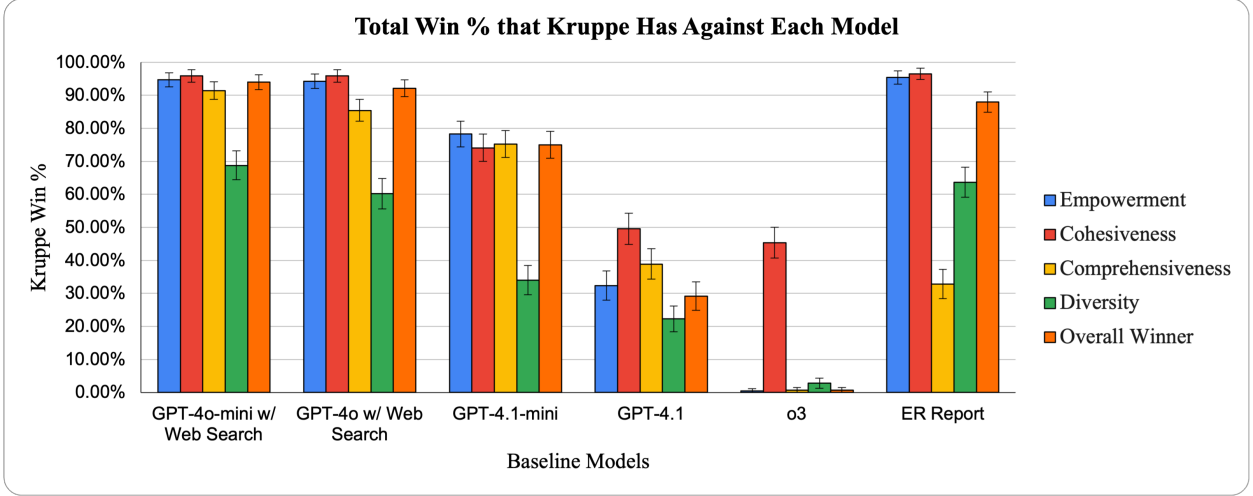


Figure 6: When comparing all KRUPPE reports, KRUPPE outperforms smaller model, but most individual reports do not outperform the more advanced models

Metrics	GPT-4o mini Web Search	GPT-4o w/ Web Search	GPT-4.1 mini	GPT-4.1	o3	ER Report
Empowerment	***	***	***			***
Cohesiveness	***	***	***			***
Comprehensiveness	***	***	***			
Diversity	***	***				***
Overall Winner	***	***	***			***

Table 1: Statistically significant KRUPPE **Total win** results. Significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

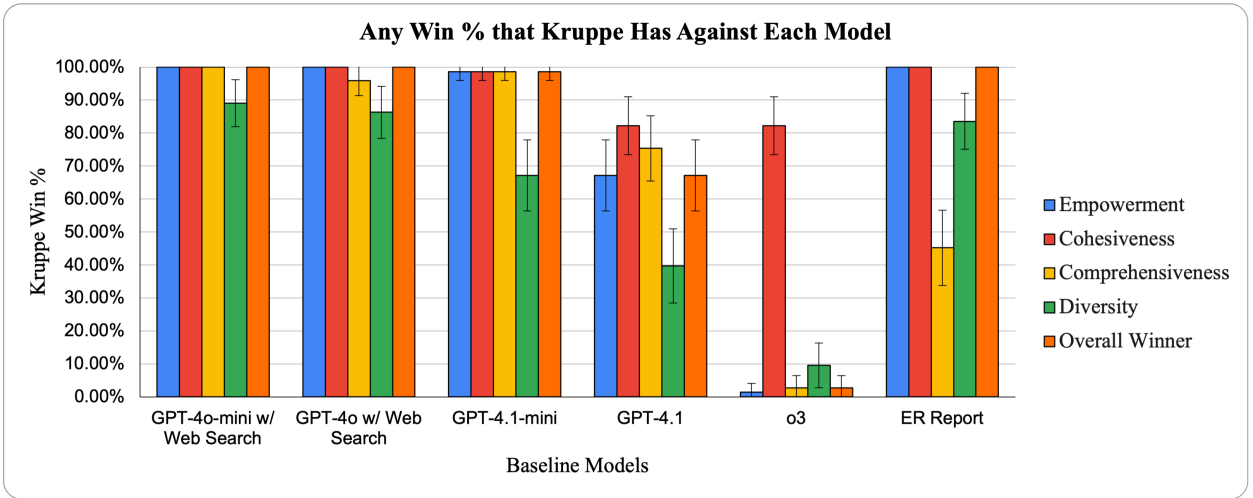


Figure 7: When evaluating reports as a set—counting a win if *any* report outperforms the baseline—KRUPPE performs significantly better. Broader exploration increases the chances that at least one report in the set captures the right narrative.

Metrics	GPT-4o mini Web Search	GPT-4o w/ Web Search	GPT-4.1 mini	GPT-4.1	o3	ER Report
Empowerment	***	***	***	**		***
Cohesiveness	***	***	***	***	***	***
Comprehensiveness	***	***	***	***		
Diversity	***	***	**			***
Overall Winner	***	***	***	**		***

Table 2: Statistically significant KRUPPE Report **Any-win** results. Significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Web Search vs. Guided Research: Compared to GPT models powered by web search, KRUPPE consistently outperforms across most metrics. Even in the stricter total-win setting, KRUPPE wins over 90% of the time on all metrics except Diversity. In the any-win setting, it wins nearly 100% of the time, again with Diversity as the only exception. This result is expected: a guided research process leads to more cohesive and analytical reports than simply feeding search results into the LLM context window.

Total-Win vs. Any-Win: Comparing Figure 6 and Figure 7, KRUPPE’s performance improves significantly in the any-win setting, where a win is recorded if *any* report from the set outperforms the baseline. Against smaller models (GPT-4.1-mini and below), KRUPPE achieves near 100% win rates across all metrics except Diversity. Even against a larger model like GPT-4.1, win rates for all metrics more than double, with all but Diversity rising to over 60% when moving from a total-win to any-win setting. However, in all cases, the model does not outperform o-3 model other than in cohesiveness.

Cohesiveness: KRUPPE reports performed strongest on the Cohesiveness metric, outperforming all models. In the any-win setting, KRUPPE achieved a 100% win rate against all models except GPT-4.1 and o-3, where it still achieved more than 80%. This result is expected, because KRUPPE’s hypothesis-driven research ensures that retrieved information contributes to a central narrative.

Diversity: KRUPPE scored lowest on the Diversity metric, with only a 39.73% win rate against GPT-4.1 in the any-win setting. Overall, KRUPPE never won more than 50% in any setting against any baseline model. This is expected, as each report focuses on a single hypothesis and lacks multiple perspectives. Although the full ensemble captures diversity, the head-to-head evaluation

compares individual reports, so this broader effect isn't reflected.

KRUPPE vs. Human Report: Surprisingly, KRUPPE performed strongly against the original equity research reports, achieving a 100% win rate in Empowerment, Cohesiveness, and Overall Winner under the any-win setting. This result is likely due to LLM evaluation bias, as the model may interpret the more technical language, graphs, and data in equity research reports as less empowering or cohesive (See appendix 7.2).

4.5 Human Feedback

In addition to the LLM-based evaluations, I collected qualitative feedback from financial analysts and investors who directly used KRUPPE. Their responses generally fell into three main themes.

Cohesive Narrative: Analysts consistently praised KRUPPE's ability to produce cohesive, narratively strong reports. They noted that traditional GPT models often simply regurgitate or summarize content, whereas KRUPPE offers meaningful analysis and presents information in a way that is much more interpretable and useful. This feedback matches aligns with the findings from the LLM-based evaluations.

Trust Issue: Despite the narrative quality, trust remains a significant concern. Analysts expressed uncertainty about that how KRUPPE generates or sources the specific information it references—such as financial figures or news events—and how it reasons to its final conclusion. This lack of trust limits KRUPPE's actionability for analysts. I believe this concern reflects broader challenge faced by all attempts to integrate LLMs into financial analysis workflows.

Understanding the Core Story: Some users observed that while KRUPPE generally answered business questions accurately (see Appendix 7.1 for an example), some reports fall short because the hypothesis failed to capture the core business story. In particular, analysts pointed out cases where KRUPPE misunderstood how a firm generates revenue, which caused the entire hypothesis—and subsequent analysis built on top of it—to fail.

Overall Ranking: Finally, one analyst indicated that they preferred the o-3 model's output compared to KRUPPE's, but ranked KRUPPE's output above all remaining GPT models. This observation matches my findings in the LLM-based evaluations.

5 Discussion and Future Work

Capturing the Business Story: Results show that KRUPPE’s hypothesis-driven approach produces reports with strong, cohesive narratives. Analyst feedback also expressed preference for the cohesive narratives, and observed that KRUPPE is able to effectively answer most of their business questions. However, some reports fall short because they failed to capture the actual business story. While KRUPPE addresses this challenge by generating multiple hypotheses from a range of perspectives, without a strong mechanism to understand core business narrative, it may still struggle to find the right conclusion. Future work should focus on improving model’s ability to identify the business story. Particularly, the lead generation stage can be improved to go beyond hypothesis generation; instead explicitly focus on uncovering a *firm’s underlying narrative*.

Human Evaluation While the LLM-based evaluation is able to evaluate KRUPPE’s reports narrative quality, it does not measure narrative *accuracy*. This is a key limitation, since a well-written report does not necessarily mean it captures real world dynamics accurately. Future work should address this limitation by involving more financial analysts to evaluate whether the hypotheses accurately captured real business stories.

Balancing Coverage and Precision: A key finding from this research is that expanding the range of hypotheses explored—especially from multiple perspectives—can significantly boost performance, even with smaller models. Although many generated reports are weak individually, grouping them together increases coverage, or the chance of uncovering a strong, relevant narrative that outperforms larger models. However, this breadth introduces a precision problem: as more hypotheses are explored, identifying the few high-quality reports among many weaker ones becomes more difficult. Future work should address this challenge by adding a final stage that identifies the strongest hypotheses and consolidates them. This process should ideally group similar narratives without flattening them into a single, generic summary.

Research-Tree Limitations: The Research-Tree framework creates multiple reasoning paths before choosing the most promising one. However, results show that *backtracking*, a key Research-Tree feature, rarely occurs in practice. This suggests the model tends to justify its initial reasoning path, even when it cannot find sufficient information to justify it. Future work should address this limitation by refining prompts to actively encourage backtracking, or by introducing early stopping

and path pruning mechanisms to push the model towards exploring alternative directions before consolidating findings into a final output.

6 Conclusion

This paper presents KRUPPE, a framework that enables LLMs to conduct financial research through a three-phase process: background research, lead generation, and hypothesis exploration. This is carried out three agents: LIBRARIAN, COORDINATOR, and HYPOTHESIS RESEARCHER agents. Experimental results show that KRUPPE can generate high-quality narratives using a smaller model, especially when outputs are evaluated as a set. In other words, as KRUPPE explores more hypotheses, the chance that at least one report captures the core business story increases.

However, KRUPPE has several limitations. The current evaluation does not assess the accuracy of the reports, and broader hypothesis coverage makes it harder to identify the strongest report. Future work should focus on addressing these issues, with the long-term goal of building a system that can out perform human analysts. The system should be able to identify the true business story behind a financial query and clearly explaining events through that central narrative.

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7 Appendix

7.1 Example Research Flow

7.1.1 Query

What economic effects would the U.S. face if China responds to Trump’s tariffs? How could this impact different U.S. stock market sectors and influence Federal Reserve policy?

7.1.2 Librarian Thought Process

[assistant] Thought 1: The research question focuses on the economic impact on the U.S. from China’s retaliation against tariffs imposed by Donald Trump, ...

Action 1: Query for background information on the economic impact of China’s retaliation against U.S. tariffs imposed by Donald Trump, focusing on overall economic effects, sectors affected, ...

[tool call] rag_query (“query”:“economic impact on the United States if China retaliates against tariffs imposed by Donald Trump, sector effects, Federal Reserve implications”, “end_date”:null, “start_date”:null)

[assistant] If China retaliates against the tariffs imposed by Trump, the U.S. economy would likely face slower growth and higher consumer prices. Analysts estimate tariffs push up costs for consumers and businesses, potentially contracting the economy by about 1% in a quarter and raising ...

[assistant] Thought 2: I now have an overview of the economic impact, sectors affected, and Federal Reserve implications. Next, identify the key U.S. stock market sectors likely most affected ...

Action 2: Query for the U.S. stock market sectors most affected by trade tensions and retaliatory tariffs from China, including major firms in these sectors.

...

7.1.3 Librarian Background Research

Final Background Report: Economic Impact on the U.S. if China Retaliates Against Tariffs Imposed by Donald Trump; Sector and Stock Market Impacts; Federal Reserve Implications

1. Overview of Economic Impact If China retaliates against the tariffs imposed by the Trump administration, the U.S. economy is likely to experience slower growth and higher consumer prices. Multiple analyses estimate that tariffs increase costs for consumers and businesses, potentially ...

2. Affected Stock Market Sectors and Key Firms The U.S. stock market sectors most affected by retaliatory tariffs from China include: ...

3. Recent News and Market Impacts Recent news highlight concerns of higher costs and layoffs in aerospace, defense, and luxury automotive sectors due to tariffs and trade tensions ...

****Conclusion:**** China retaliating against U.S. tariffs would likely slow U.S. economic growth by raising costs, disrupting trade, and depressing exports. The aerospace, defense, automotive, and consumer goods sectors would be majorly affected, impacting major U.S. stock market companies. Market volatility may increase while some firms face earnings pressures. The Federal Reserve would need to navigate a complex environment, balancing inflation control with growth concerns, likely complicating monetary easing efforts.

...

7.1.4 Generated Experts

International Trade Economist: The International Trade Economist specializes in analyzing global trade policies, tariff impacts, and economic relations between countries

Geopolitical Risk Analyst: The Geopolitical Risk Analyst studies how international tensions and government actions affect economic stability, market risk perceptions

...

7.1.5 Final Outputs

Output 1/6:

Expert Perspective:

Monetary Policy Specialist

Expert Description:

The Monetary Policy Specialist is an expert in central banking and Federal Reserve operations, capable of assessing how trade-induced economic shocks might steer policy decisions on rate adjustments, quantitative easing, and inflation targeting to stabilize the U.S. economy.

Report:

Working Hypothesis (Fully Developed): China's retaliation against tariffs imposed by the Trump administration has caused a modest but meaningful slowdown in U.S. economic growth accompanied by inflationary pressures driven by higher input and consumer costs. These effects disproportionately burden trade-exposed sectors—including aerospace, automotive, energy, and technology—leading to increased earnings volatility and negative stock market sentiment. In response, the Federal Reserve has adopted a cautious, data-dependent monetary policy stance, maintaining relatively high interest rates and avoiding aggressive adjustments until inflation moderates and trade uncertainties abate.

Final Research Report:

The imposition of tariffs by the Trump administration and China's subsequent retaliatory tariffs have materially affected the U.S. economy and financial markets. Empirical analysis reveals that China's tariffs significantly strained sectors reliant on international trade and supply chains, including aerospace (notably Boeing), energy, telecommunications, automotive, and homebuilding. These sectors experienced disrupted supply chains, lost export sales, and elevated costs, which dampened corporate profits and contributed to investor selloffs and stock market volatility. This sector-specific impact confirms that the economic consequences of the trade war were heterogeneous, with trade-exposed industries bearing the brunt of financial market risks.

Cost increases stemming from tariff passthrough elevated inflationary pressures. Fed Chair Jerome Powell and other Federal Reserve officials have acknowledged the risk of tariffs generating at least

temporary inflation increases, with the persistence of these effects contingent upon demand dynamics. In this environment, inflation pressures contributed to a complex tradeoff with decelerating economic growth. Some demand moderation likely tempered inflation persistence, but cost shocks nonetheless pushed consumer prices and input costs upward, particularly in affected sectors.

The Federal Reserve reacted with caution and prudence. Rather than aggressively cutting or raising interest rates, the Fed adopted a wait-and-see approach, carefully balancing the risks of inflation against the threat of slowing growth. Historical analogs demonstrate that in comparable trade shock episodes, the Fed prioritizes anchoring long-term inflation expectations and controls inflation pressures without prematurely loosening monetary policy. This careful stance reflects the significant uncertainty in trade policy outcomes and the Fed’s objective to avoid fueling entrenched inflation or tipping the economy into recession.

Stock market performance mirrored these economic realities. Earnings volatility increased substantially in trade-sensitive sectors; companies reliant on global supply chains, international markets, and tariff-impacted inputs saw heightened uncertainty and downward pricing pressure. Investors rotated away from vulnerable sectors and U.S. assets toward global alternatives, amplifying financial market volatility. Technology firms with significant exposure to Chinese components faced greater shocks beyond pure tariff rates. These developments underscore the multifaceted nature of trade retaliation impact, through cost structures, demand effects, and sentiment swings.

Overall, China’s retaliatory tariffs prompted a modest but significant slowdown in U.S. growth, fueled inflation pressures, and created heterogeneous sector effects in the stock market. The Federal Reserve’s stance of cautious monetary policy—retaining relatively high interest rates and avoiding abrupt moves—reflects a balanced response to inflation-growth tradeoffs and substantial trade policy uncertainty. This integrated view blends macroeconomic, sectoral, and monetary policy dimensions to illuminate the complex dynamics of trade-induced macro-financial shocks and policy reactions.

This comprehensive synthesis supports the fully developed working hypothesis that China’s tariff retaliation moderates U.S. growth and elevates inflation costs, applies uneven burdens across sectors generating financial market volatility, and prompts the Federal Reserve to maintain prudent, measured monetary policy until clearer inflation moderation and trade resolution emerge. This hypothesis can inform future monitoring of evolving trade conflicts, sector risk exposures, inflation

trends, and central bank policy pathways.

Action: FINISH[complete]

Output 2/6:

Expert Perspective:

Monetary Policy Specialist

Expert Description:

The Monetary Policy Specialist is an expert in central banking and Federal Reserve operations, capable of assessing how trade-induced economic shocks might steer policy decisions on rate adjustments, quantitative easing, and inflation targeting to stabilize the U.S. economy. Report: Final Hypothesis:

China's retaliation against tariffs imposed by the Trump administration generated a supply-side shock that elevated costs for U.S. producers and consumers, leading to moderate inflation and dampened economic growth. This impact was particularly severe in trade-exposed sectors such as aerospace, automotive, and consumer goods, which faced increased production costs, disrupted supply chains, and reduced export opportunities. The resulting sector-specific financial pressures and uncertainty heightened volatility within U.S. stock markets. In response to these trade tensions and the inflation-growth trade-off, the Federal Reserve adopted a cautious, more accommodative monetary policy stance, including interest rate cuts, to mitigate downside growth risks while balancing inflationary pressures.

Final Research Report:

This analysis sought to understand how Chinese retaliatory tariffs on U.S. goods, imposed during the Trump administration's trade conflict, affected the U.S. economy, specific industry sectors, the stock market, and Federal Reserve policy decisions. Despite limited direct empirical documents available in the internal vector database, a thorough synthesis based on economic theory, historical precedent, and known market and policy outcomes was conducted.

The trade conflict initiated significant supply-side shocks in the U.S., raising input and intermediate goods costs due to tariffs on key raw materials and components. The aerospace sector, for instance, experienced increased costs on critical inputs such as titanium, while simultaneously facing reduced

sales to China—one of the fastest-growing markets—due to retaliatory tariffs. Competitive pressure against firms like Boeing intensified, squeezing profit margins and dampening future growth prospects.

Similarly, the automotive sector encountered higher manufacturing costs through tariffs on parts and components embedded in complex global supply chains. These cost increases cascaded to consumer prices, suppressing demand both domestically and abroad, where China’s tariffs curtailed vehicle exports. The consumer goods sector was not spared, facing tariff-induced price hikes on both final products and inputs, leading to inflationary pressures that affected consumer purchasing behavior and broader cost-push inflation in the economy.

Stock market behavior during this period mirrored these underlying economic stresses. Equity volatility surged, particularly in sectors directly exposed to tariffs such as technology, industrials, and consumer discretionary segments. Investor risk appetite fluctuated in response to trade war news, with affected sectors often experiencing underperformance due to concerns over margin compression and uncertainties about supply chain continuity. Meanwhile, more domestic-focused firms occasionally outperformed, illustrating divergent stock market impacts across sectors.

The Federal Reserve closely monitored these developments as part of the broader macroeconomic environment. Tariffs contributed to cost-push inflation while simultaneously acting as a growth dampener by creating uncertainty and reducing business investment. This complex scenario challenged the Fed’s dual mandate to promote maximum employment and price stability in the face of conflicting signals. Starting in mid-2019, as trade tensions persisted and global growth softened, the Fed moved toward a more accommodative stance, cutting interest rates multiple times to support economic expansion and offset tariff-related headwinds.

In summary, Chinese retaliatory tariffs imposed during the Trump administration significantly affected the U.S. economy through increased costs and supply chain disruptions, notably in key trade-exposed sectors. These impacts fueled stock market volatility and complicated Federal Reserve policy, which ultimately adopted a cautious and supportive approach to maintain growth amid tariff-driven inflationary pressures. This narrative confirms the preliminary hypothesis, underscoring the intricate interplay between trade policy shocks and macroeconomic and financial market dynamics in the U.S. during this period.

Output 3/6:

Expert Perspective:

International Trade Economist

Expert Description:

The International Trade Economist specializes in analyzing global trade policies, tariff impacts, and economic relations between countries, particularly between the U.S. and China. They provide insights into how retaliatory tariffs influence trade flows, manufacturing, exports/imports, and overall economic output.

Report:

Final Research Report on the Economic Impact of China's Retaliation Against U.S. Tariffs

Final Hypothesis: China's retaliation to U.S. tariffs imposed during the Trump administration generated notable supply-side shocks that increased input costs and disrupted export flows, disproportionately hitting trade-exposed U.S. sectors such as automotive and aerospace. Consumer goods sectors showed resilience by passing inflationary costs to consumers without major demand loss. The defense sector, supported by strong government contracts and strategic reshoring efforts, demonstrated robust corporate and stock market performance amid this uncertainty. Despite inflationary pressures partly driven by tariffs, the Federal Reserve prioritized inflation control through sustained monetary tightening, navigating a complex trade-off between curbing inflation and sustaining economic growth. This policy stance reflects the constrained environment hypothesized, where tariff-driven inflation limits Fed easing despite slowing growth.

Narrative Summary:

1. Economic and Sectoral Impact of China's Retaliation to U.S. Tariffs The imposition of tariffs by the Trump administration and subsequent retaliatory tariffs by China led to increased production costs and disrupted supply chains in key U.S. trade-dependent sectors. Evidence from automotive and aerospace industries shows rising costs and uncertainty: major firms like Mercedes-Benz and GM withdrew earnings guidance amid tariff-driven challenges, while aerospace producers raised prices to offset increased input costs. Consumer electronics partially evaded tariffs through exemptions, yet overall uncertainty persisted. These supply disruptions validate the hypothesis that trade-exposed sectors faced slowed growth and profitability declines.

2. **Sectoral Heterogeneity and Corporate Responses** In contrast to trade-sensitive sectors, the defense industry stood out as resilient due to large, stable government contracts and reshoring and diversification strategies encouraged by multiyear spending commitments. Defense contractors such as Lockheed Martin and Raytheon experienced strong stock price gains in the 2018–2024 period, buoyed also by geopolitical events like increased arms orders related to the Ukraine conflict. Consumer goods companies showed the ability to raise prices (e.g., Starbucks, McDonald’s) without reducing demand, maintaining revenue resilience but contributing to sustained inflationary pressure—this partial price inelasticity among consumers moderated some negative growth impacts but reinforced inflation dynamics.

3. **Financial Market Reactions** While defense and certain consumer-oriented stocks performed positively or showed resilience, the automotive sector’s stock market effects directly attributable to tariffs remain less documented in available data. Tariff-related uncertainty contributed to elevated market volatility. The heterogeneous sector outcomes align with the hypothesis that some areas of the U.S. economy suffered greater tariff-induced shocks, whereas others compensated through stable contracts or consumer pricing strategies.

4. **Federal Reserve Policy Considerations** Federal Reserve officials, including Chair Jerome Powell, consistently identified inflation as a principal threat during the tariff period, with tariffs contributing as part of broader supply chain and geopolitical inflation drivers. The Fed implemented a series of interest rate increases to counteract inflation, noting the risk of entrenched inflation as intolerable despite the associated risks of slowing growth or recession. Fed communications around 2018–2024 emphasize cautious but decisive tightening, underscoring the policy dilemma that the working hypothesis anticipated: tariff-driven inflation constrains monetary easing even when growth pressures mount. Forecasts toward 2024–2025 suggest inflation is expected to ease below the Fed’s 2

5. **Limits and Gaps in Data** Direct, sector-specific quantitative corporate earnings data specifically linked to tariffs was not found. Likewise, detailed stock market performance for the automotive sector during the tariff period was limited. However, available evidence from corporate announcements, broader sectoral analyses, and Federal Reserve policy signals effectively illustrates the overall narrative that tariff retaliation caused differentiated sectoral economic effects and complex monetary policy trade-offs.

Conclusion: The evidence consolidates the hypothesis that China’s retaliatory tariffs imposed a

supply-side shock that suppressed growth in trade-dependent U.S. sectors by raising costs and disrupting exports while sparing sectors supported by government contracts or consumer price insensitivity. The Federal Reserve responded to the inflationary pressures, including tariffs, with sustained interest rate hikes aimed at preventing inflation entrenchment despite economic slowdown risks. This confluence confirms the working hypothesis and highlights the multi-faceted impact of tariff retaliation on the U.S. economy, stock market sectors, and monetary policy.

Recommendations for Further Research: Future analyses could benefit from more granular corporate earnings and stock return data to quantify tariff impacts by sector precisely. Investigation into longer-term structural responses, such as supply chain diversification and reshoring outcomes, can further clarify corporate resilience patterns. Additionally, examining consumer and investor sentiment evolution amid tariff uncertainty might yield deeper insights into behavioral adaptations shaping inflation and market dynamics.

End of Report.

Output 4/6: Expert Perspective:

International Trade Economist

Expert Description:

The International Trade Economist specializes in analyzing global trade policies, tariff impacts, and economic relations between countries, particularly between the U.S. and China. They provide insights into how retaliatory tariffs influence trade flows, manufacturing, exports/imports, and overall economic output.

Report:

Final Hypothesis:

China's retaliatory tariffs against U.S. tariffs imposed during the Trump administration caused a modest but meaningful drag on U.S. economic growth, slowing GDP expansion by approximately 0.1 to 0.3 percentage points annually rather than the initially hypothesized 1% quarterly contraction. The economic impact was uneven across sectors, with manufacturing, aerospace, automotive, and consumer goods bearing significant cost pressures and earnings volatility. Commercial aerospace firms like Boeing faced notable vulnerabilities due to trade retaliation and supply chain disruptions, whereas defense firms demonstrated relative resilience owing to steady government contracting and

limited import exposure. Firms across impacted sectors mitigated tariff effects through supply chain diversification, reshoring, and inventory management. The Federal Reserve responded to the tariffs' inflationary cost pressures and growth-slowing effects with a cautiously accommodative monetary policy stance, employing interest rate reductions to sustain economic momentum amid uncertainty.

Final Research Report:

The imposition of tariffs by the Trump administration on Chinese imports, and China's subsequent retaliatory tariffs on U.S. goods, constituted significant disruptions in global trade between the world's two largest economies during 2018-2020. The economic effects of this trade conflict were multi-dimensional, affecting GDP growth, sectoral financial performance, consumer prices, and monetary policy.

Macroeconomically, the tariff-induced cost shocks raised input costs across U.S. industries dependent on Chinese intermediate goods, thereby increasing producer and consumer prices. Multiple analyses, including those from the Federal Reserve and independent economic studies, estimate that the cumulative impact of these tariffs reduced U.S. GDP growth by around 0.1 to 0.3 percentage points per year during peak tariff periods. This drag was driven by heightened production costs and increased uncertainty, which dampened business investment and complicated supply chains. The initially posited 1% quarterly contraction was not supported by available evidence; instead, the effect, while persistent, was quantitatively more moderate.

Sectoral impacts revealed heterogeneous vulnerabilities. Manufacturing sectors, reliant on imported components, encountered compressed profit margins and interruptions, accelerating efforts to diversify supply chains beyond China. Aerospace companies, particularly Boeing, incurred cost pressures from tariffs on aluminum and aerospace parts and faced challenges from China's retaliatory aircraft export tariffs. However, defense-related aerospace segments showed more resilience due to government contracts and a lower reliance on tariffed imports. Automotive manufacturers contended with higher costs for parts and raw materials, leading to earnings volatility and mixed stock market performances. Similarly, consumer goods companies experienced uneven earnings impacts, balancing cost absorption and price increases, with premium brands better positioned to maintain margins. The defense sector exhibited relative stability owing to steady government demand and limited tariff exposure.

Firm-level adaptive responses were notable across sectors: supply chain diversification to non-Chinese sources (including countries like Vietnam and Mexico), reshoring production facilities, inventory optimization, and investments in supply chain technology enhanced resilience and mitigated some tariff costs. These strategies contributed to stabilizing earnings and stock valuations amid tariff-related pressures.

On the monetary policy front, the Federal Reserve confronted a complex environment: tariffs exerted upward inflationary pressures by increasing input prices, a form of cost-push inflation distinct from demand-driven inflation, while simultaneously weighing the slowing effect tariffs imposed on overall economic growth. This inflation-growth tradeoff required a delicate approach. Between 2018 and 2019, amid global growth headwinds exacerbated partly by trade tensions, the Federal Reserve progressively lowered interest rates three times and maintained a more accommodative stance than it would have absent tariffs. The objective was to support economic expansion without allowing inflation expectations to destabilize, reflecting the tariff-induced policy ambiguity anticipated in the original hypothesis.

In summary, the multifaceted economic impact of China's retaliation on U.S. tariffs was significant though less severe than initially estimated. The nuanced effects ranged from measurable GDP slowdown, sector-specific cost and earnings pressures, strategic corporate adaptation, to constrained but responsive monetary policy. This comprehensive assessment highlights the interconnectedness of global trade and the challenges facing policymakers balancing inflation and growth in the face of protectionist shocks.

This evidence supports the refined hypothesis, providing a balanced and empirically grounded understanding of the economic repercussions of China's retaliatory tariffs on the United States.

Output 5/6: Expert Perspective:

Equity Market Sector Analyst

Expert Description:

The Equity Market Sector Analyst has expertise in stock market dynamics and sector performance, interpreting how geopolitical events and tariffs influence investor sentiment, valuations, and capital flows within distinct sectors like technology, agriculture, industrials, and consumer discretionary.

Report:

Final Hypothesis: If China retaliates against U.S. tariffs, the United States will experience a short-term GDP growth slowdown of approximately 1% accompanied by increased inflationary pressures due to higher input costs and supply chain disruptions. This will disproportionately impact trade-exposed sectors such as aerospace, automotive, transportation, energy, telecommunications, homebuilding, and consumer goods, leading to heightened stock market sector volatility and differentiated firm performance—financially robust companies showing resilience while vulnerable firms face earnings pressure and valuation declines. The Federal Reserve will confront a challenging policy environment that balances controlling inflation caused by tariff-induced cost shocks against slowing growth, likely resulting in prolonged constrained monetary easing and sustained elevated interest rates. Supply chain adaptations and tariff exemptions will provide partial, short-term relief but will not fully mitigate the economic and market impacts. Meanwhile, increased market volatility and investor risk aversion will further complicate capital flows and sector valuations in the equity markets.

Final Research Report: The economic consequences of a Chinese retaliation to U.S. tariffs reflect a multifaceted challenge to the United States economy, the equity markets, and Federal Reserve monetary policy. Analysis of economist reports and market observations consistently found that such retaliation results in higher costs for both U.S. businesses and consumers. Tariffs disrupt global supply chains and increase prices, which squeeze real wages and consumer spending, thereby slowing U.S. GDP growth by about 1% in the short term. These dynamics raise inflationary pressures, contributing to broader economic risks including an elevated recession threat. Although direct quantitative models of the specific magnitude of China's retaliation effects are sparse, consensus views align with meaningful negative impacts on U.S. economic activity.

Sectoral impacts are uneven but material. Key trade-exposed sectors such as aerospace, automotive, transportation, energy, telecommunications, homebuilding, and consumer goods bear the brunt of tariff-related pressures. For instance, firms like Colgate-Palmolive have cited tariff-related cost increases of approximately 2.5% of their cost of goods sold, while companies in aerospace and transportation report supply chain disruptions and demand uncertainties. Over 90% of S&P 500 companies flagged tariffs as a significant operational challenge, and 44% voiced concerns about recession risks at earnings calls, underscoring widespread market unease. This drives amplified earnings volatility and valuation pressure in affected sectors, contributing to overall stock market turbulence.

Firm-level financial health plays a moderating role. Robust, financially healthy companies employing adaptive strategies—such as restructuring supply chains and shifting production outside China—show greater resilience to tariff shocks. However, many firms still face uncertainty, causing cautious earnings forecasts and intermittent stock price declines linked to trade tensions. State interventions in China also help stabilize some firms exposed to the trade environment, but overall vulnerability remains highly heterogeneous. This differentiation at the firm level further contributes to sectoral and market volatility.

In response to inflation pressures fueled by tariff-induced cost shocks and economic slowdown risks, the Federal Reserve has adopted a cautious and complex policy stance. It has prioritized anchoring inflation expectations by maintaining elevated interest rates despite weakening growth prospects. The Fed refrains from premature easing and closely monitors incoming data, an approach framed as a balancing act between its dual mandate of price stability and maximum employment. This posture is expected to continue, with restrained monetary easing and potentially prolonged tight policy likely, to curb inflation without precipitating greater economic downturn.

Mitigation efforts provide some relief but not full protection. U.S. and Chinese authorities have granted exemptions on tariffs for critical goods such as healthcare products, semiconductors, medical equipment, and aviation parts, alleviating some cost pressures temporarily. Supply chain adaptations including inventory stockpiling, rerouting goods via third countries, and warehousing strategies have helped businesses navigate tariff obstacles. Yet, prolonged tariffs pose risks of escalating trade disruptions and inflation that could outpace these measures.

The trade tensions have also significantly influenced investor behavior and financial markets. The escalating tariff conflict drove increased market volatility and elevated risk aversion, with global stock markets undergoing sharp asset swings. Investors have notably reduced U.S. equity holdings amid recession fears and uncertainty about trade outcomes, while safe-haven assets like gold experienced significant price rallies. Concerns about China potentially offloading U.S. Treasury holdings have added to market unease. These developments magnify volatility particularly in sectors vulnerable to tariffs, affecting capital allocation and valuation dynamics.

In summary, the evidence strongly supports the hypothesis that Chinese retaliation escalates cost and supply-side shocks that meaningfully slow U.S. growth and raise inflation in the short term. The impacts are sector-specific and moderated by firm-level financial strength, yet uncertainty and

cost pressures remain pervasive. The Federal Reserve's response reflects a constrained monetary policy environment aimed at balancing inflation containment with growth risks. Supply chain adaptations and tariff exemptions offer partial mitigation but do not undo the broader economic and market effects. Heightened volatility and shifting investor sentiment further complicate the equity market landscape, reinforcing the interconnected challenges posed by tariff retaliation.

This comprehensive evaluation underscores that the economic, equity market, and monetary policy consequences of Chinese retaliation against U.S. tariffs comprise a significant and sustained challenge to the U.S. economy that demands nuanced sector-aware investment strategies and careful central bank policy calibration.

Output 6/6:

Expert Perspective:

Equity Market Sector Analyst

Expert Description:

The Equity Market Sector Analyst has expertise in stock market dynamics and sector performance, interpreting how geopolitical events and tariffs influence investor sentiment, valuations, and capital flows within distinct sectors like technology, agriculture, industrials, and consumer discretionary.

Report:

Final Working Hypothesis:

If China retaliates against U.S. tariffs, the resulting economic impact on the United States will include significantly slowed GDP growth and increased inflation driven by higher input costs and supply chain disruptions. Trade-exposed sectors—such as aerospace, automotive, and consumer goods—will experience notable earnings pressure and heightened stock price volatility. Defense sectors, supported by relatively stable government contracts, will demonstrate more resilience and partially offset losses faced by other industries. This combination of inflationary pressures amid economic slowdown will constrain the Federal Reserve's policy flexibility, leading to a cautious or tighter monetary stance aimed at containing inflation without precipitating a recession.

Final Research Report:

This research explored the economic consequences of potential Chinese retaliation against U.S.

tariffs, focusing on macroeconomic outcomes, stock market sector effects, and Federal Reserve policy implications. Although direct empirical datasets were initially limited, comprehensive insights were obtained primarily through recent retrospective news analyses and expert commentaries.

China's retaliatory tariffs, some reaching up to 125%, coupled with U.S. rates as high as 145% during escalations, have materially impacted the U.S. economy. Notably, Goldman Sachs revised U.S. GDP growth forecasts down sharply from 2.8% to 0.5%, citing tariff-induced financial tightening and policy uncertainty. Inflation pressures increased as tariffs boosted import costs on key goods, including electronics, machinery, textiles, and furniture. This translated into higher input prices for businesses and elevated consumer prices, fulfilling the hypothesis that tariffs contribute to inflation via cost-push mechanisms.

Sectoral impacts were pronounced and aligned with expectations. Aerospace companies like Boeing endured supply chain disruptions and cost inflation that suppressed earnings and led to volatile stock prices. The automotive sector faced a recession-like demand environment with tariffs on vehicles and parts, forcing cost-cutting measures. Consumer goods companies, including retailers such as Walmart, Target, and Procter & Gamble, experienced consumer pullbacks attributed to tariff fears and price uncertainty. While defense sectors confronted export uncertainties, their dependence on stable government contracts conferred relative resilience compared to more trade-exposed industries. Earnings revisions across these trade-exposed sectors were predominantly downward, reflecting risk-averse corporate outlooks amid tariff uncertainty.

Federal Reserve Chair Jerome Powell characterized tariffs as a significant policy challenge, as they exacerbate inflation while simultaneously slowing growth—creating a “challenging scenario” for monetary policy. The Fed's emphasis was on preventing inflation from becoming entrenched, which limited their capacity to ease monetary policy despite growth headwinds. This stance was reflected in cautious interest rate adjustments: rate hikes were initiated during rising inflation but tempered after economic slowdowns, including significant cuts in 2020 responding to the pandemic-driven downturn. Post-pandemic monetary policy continued to balance inflation and growth concerns with a relatively tighter posture compared to peer central banks. While tariffs contributed “noise” to economic forecasts, they formed part of a complex mix pressuring inflation and dampening growth expectations, thus restraining the Fed's flexibility.

Stock market volatility during tariff escalations was marked, driven by investor concerns over supply

chain disruptions, rising costs, and uncertain corporate earnings. Trade-exposed sectors displayed heightened price swings and valuation adjustments as market participants recalibrated risk premia. Defense and government-contracted sectors showed more muted stock price reactions, consistent with their defensive characteristics.

In conclusion, the hypothesis that Chinese retaliation against U.S. tariffs materially slows GDP growth, fuels inflation, imposes earnings pressures and volatility on trade-exposed sectors, with defense showing resilience, and constrains Federal Reserve monetary policy is confirmed through multiple lines of evidence. This multifaceted impact paints a challenging macroeconomic environment where inflation control and growth management must be delicately balanced amid ongoing geopolitical trade tensions.

This analysis highlights the importance for investors and policymakers to monitor tariff developments closely, considering their broad-based impacts on the economy, sector earnings, stock valuations, and monetary policy decisions.

7.2 KRUPPE vs. Equity Research Report

The following is a summary of the model’s justification for preferring KRUPPE or the ER Reports on the four metrics. Answer 1 refers to KRUPPE reports, and Answer 2 refers to the ER Reports.

Empowerment: “Answer 1 consistently empowers readers by guiding them through the logic, context, and implications of each issue; it translates facts into understanding, explicitly relates risks and opportunities, and equips a broad audience—not just experts—with the tools and confidence to make informed, independent judgments. In contrast, *Answer 2 is typically more technical and data-heavy, often overwhelming non-specialist readers and providing less interpretive guidance, context, or empowerment.*”

Cohesiveness: “Answer 1 consistently outperformed Answer 2 because it maintained a clear, logical, and unified narrative—structuring its analysis around a central hypothesis, building arguments step by step, and linking each section back to the overarching thesis. . . In contrast, *Answer 2 was frequently fragmented, presented as a collection of disjointed data points, tables, analyst notes, and disclosures, which disrupted narrative flow and made it difficult for readers to follow a single, coherent story.*”

Comprehensiveness: “The qualities that made *Answer 2 better are its exceptional breadth and depth of quantitative detail*, including extensive financial data, granular forecasts, scenario analyses, segment breakdowns, risk disclosures, and explicit modeling—offering a near-exhaustive coverage of all relevant facts and data points. . .”

Diversity: “Answer 1 is consistently better because it draws from a broader and more diverse range of perspectives, blending quantitative financial analysis with qualitative viewpoints such as strategy, regulatory landscapes, peer comparisons, industry trends, risks, and stakeholder implications. This . . . allows for richer insight and greater diversity in its responses compared to the *narrower, more internally or financially focused analyses often found in Answer 2.*”

7.3 Data Tables

Percentage win that KRUPPE has against each model.

Metrics	GPT 4o mini Web Search	GPT 4o w/ Web Search	GPT 4.1 mini	GPT 4.1	o3	ER Report
Empowerment	94.68%	94.21%	78.24%	32.41%	0.46%	95.37%
Cohesiveness	95.83%	95.83%	74.07%	49.54%	45.37%	96.53%
Comprehensiveness	91.44%	85.42%	75.23%	38.89%	0.69%	32.87%
Diversity	68.75%	60.19%	34.03%	22.29%	2.78%	63.66%
Overall Winner	93.98%	92.13%	75.00%	29.17%	0.69%	87.96%

Table 3: Percentage of Total KRUPPE Report Wins

Metrics	GPT 4o mini Web Search	GPT 4o w/ Web Search	GPT 4.1 mini	GPT 4.1	o3	ER Report
Empowerment	100.00%	100.00%	98.63%	67.12%	1.37%	100.00%
Cohesiveness	100.00%	100.00%	98.63%	82.19%	82.19%	100.00%
Comprehensiveness	100.00%	95.89%	98.63%	75.34%	2.74%	45.21%
Diversity	89.04%	86.30%	67.12%	39.73%	9.59%	83.56%
Overall Winner	100.00%	100.00%	98.63%	67.12%	2.74%	100.00%

Table 4: Percentage of Any KRUPPE Report Wins

Metrics	GPT-4o- mini w/ Internet	GPT-4o w/ Internet	GPT-4.1- mini	GPT-4.1	o3	Original ERPs
Empowerment	< 0.001	< 0.001	< 0.001	> 0.99	> 0.99	< 0.001
Cohesiveness	< 0.001	< 0.001	< 0.001	0.58	0.97	< 0.001
Comprehensiveness	< 0.001	< 0.001	< 0.001	> 0.99	> 0.99	> 0.99
Diversity	< 0.001	< 0.001	> 0.99	> 0.99	> 0.99	< 0.001
Overall Winner	< 0.001	< 0.001	< 0.001	> 0.99	> 0.99	< 0.001

Table 5: P-values for Total KRUPPE Win across models and metrics.

Metrics	GPT-4o- mini w/ Internet	GPT-4o w/ Internet	GPT-4.1- mini	GPT-4.1	o3	Original ERPs
Empowerment	< 0.001	< 0.001	< 0.001	0.002	> 0.99	< 0.001
Cohesiveness	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Comprehensiveness	< 0.001	< 0.001	< 0.001	< 0.001	> 0.99	0.79
Diversity	< 0.001	< 0.001	0.002	0.96	> 0.99	< 0.001
Overall Winner	< 0.001	< 0.001	< 0.001	0.002	> 0.99	< 0.001

Table 6: P-values for Any KRUPPE Win across models and metrics.